# EDA 898

Reviews of Linear Algebra and Matrix Computations

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# Prologue

In the next slides, we will review some fundamental concepts in linear algebra and matrix analysis that are essential for understanding the mathematical foundations of EDA. We will cover topics such as eigenvalues, eigenvectors, diagonalization and matrix functions as they are needed in this EDA course. These concepts will provide the necessary background for analyzing and designing complex systems in the context of electronic design automation.

# Eigenvalues, Eigenvectors, and Diagonalization

# Eigenvalues and Eigenvectors – i

**Eigenvectors and Eigenvalues:**

An **eigenvector** of an $n \times n$ matrix $\mathbf{A}$ is a nonzero vector $\mathbf{v}$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

for some scalar $\lambda$.

$\lambda$ is an **eigenvalue** of $\mathbf{A}$ if there exists a nontrivial solution $\mathbf{v}$ ($\mathbf{v} \neq \mathbf{0}$) to this equation; such a vector $\mathbf{v}$ is called an **eigenvector corresponding to** $\lambda$.

From the above definition, a scalar $\lambda$ is an eigenvalue of an $n \times n$ matrix $\mathbf{A}$ iff the homogeneous linear system

$$(\mathbf{A} - \lambda\mathbf{I})\ \mathbf{v} = \mathbf{0} \tag{1}$$

has a nontrivial solution ($\mathbf{v} \neq \mathbf{0}$). Since $\mathbf{v}$ is nonzero, the matrix $(\mathbf{A} - \lambda\mathbf{I})$ must be singular, i.e., *non-invertible*. A matrix is singular if and only if its determinant is zero; therefore, the eigenvalues of $\mathbf{A}$ are precisely the scalars $\lambda$ for which

# Eigenvalues and Eigenvectors – ii

$$\det\left(\mathbf{A} - \lambda\mathbf{I}\right) = 0\,. \tag{2}$$

**Characteristic Polynomial:**

For a square matrix $\mathbf{A}$, $\det\left(\mathbf{A} - \lambda\mathbf{I}\right)$ is a polynomial of degree $n$ called the **characteristic polynomial** of $\mathbf{A}$, i.e.,

$$p(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^n + b_{n-1}\lambda^{n-1} + \cdots + b_1\lambda + b_0.$$

**Computing Eigenvalues:**

The eigenvalues of a square matrix $\mathbf{A}$, are obtained by computing the roots of $\det\left(\mathbf{A} - \lambda\mathbf{I}\right)$. Since a degree-$n$ polynomial has exactly $n$ roots (counting multiplicities), an $n \times n$ matrix can have $n$ eigenvalues.

$$p(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^n + b_{n-1}\lambda^{n-1} + \cdots + b_1\lambda + b_0.$$

# Eigenvalues and Eigenvectors – iii

**In a more mathematical setting**

One can show that the set of eigenvalues of $\mathbf{A}$ is precisely the **spectrum** of $\mathbf{A}$, denoted $\sigma(\mathbf{A})$. An eigenvector is a dimension that remains invariant under pre-multiplication by $\mathbf{A}$. Let $\mathcal{S}$ be a subspace as $\mathcal{S} \subseteq \mathbb{C}^n$ with the property

$$\mathbf{v} \in \mathcal{S} \implies \mathbf{A}\mathbf{v} \in \mathcal{S}$$

Hence $\mathcal{S}$ is called an **invariant subspace** of $\mathbf{A}$. The importance of eigenvector is that it is spanning this invariant subspace of $\mathbf{A}$.

- If $\lambda \in \mathbb{R},\ \lambda > 0$: $\mathbf{v}$ and $\mathbf{A}\mathbf{v}$ point in the same direction.
- If $\lambda \in \mathbb{R},\ \lambda < 0$: $\mathbf{v}$ and $\mathbf{A}\mathbf{v}$ point in opposite directions.
- If $\lambda \in \mathbb{R},\ |\lambda| < 1$: $\mathbf{A}\mathbf{v}$ is shorter than $\mathbf{v}$.
- If $\lambda \in \mathbb{R},\ |\lambda| > 1$: $\mathbf{A}\mathbf{v}$ is longer than $\mathbf{v}$.

# Question

**What does it mean for a matrix $\mathbf{A}$ to have an eigenvalue equal to $0$?**

**<u>Answer:</u>**

A matrix $\mathbf{A}$ has $0$ as an eigenvalue exactly when the equation

$$\mathbf{A}\,\mathbf{v} = 0\,\mathbf{v} \tag{3}$$

admits a nontrivial solution. Since (3) reduces to

$$\mathbf{A}\,\mathbf{v} = \mathbf{0},$$

a nontrivial solution exists precisely when $\mathbf{A}$ is *not invertible*.

$$\textbf{$\mathbf{A}$ has $0$ as an eigenvalue} \quad \Longleftrightarrow \quad \textbf{$\mathbf{A}$ is singular.}$$

# Example: Eigen Decomposition of a Matrix – i

To compute eigenvalues:

$$\mathbf{A}\mathbf{v} \;=\; \lambda\mathbf{v} \quad \Longrightarrow (A - \lambda\mathbf{I})\,\mathbf{v} = \mathbf{0}$$

To have a nontrivial solution as $\mathbf{v} \neq \mathbf{0}$, it is,

$$\det\left(A - \lambda\mathbf{I}\right) = |A - \lambda\mathbf{I}| = 0$$

**Example:**

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \qquad |A - \lambda I| = 0 \;\Longrightarrow\; \lambda^2 - 5\lambda + 4 = 0 \;\Longrightarrow\; \text{Eigenvalues: } 1,\ 4$$

One can also investigate that: $\mathbf{A}^2 - 5\mathbf{A} + 4\mathbf{I} = \mathbf{0}$.

$$\mathbf{A}\mathbf{v}_1 = \lambda_1\mathbf{v}_1 \;\Longrightarrow\; \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 1 \times \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

# Example: Eigen Decomposition of a Matrix – ii

$$\begin{cases} 2v_1 + 2v_2 = 0 \\ v_1 + v_2 = 0 \end{cases} \implies v_1 + v_2 = 0 \text{ for } v_1 = 1 \implies v_2 = -1$$

$$\text{One eigen-pair is } (\lambda_1 = 1,\ \mathbf{x}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix})$$

and

$$\mathbf{A}\mathbf{v}_2 = \lambda_2 \mathbf{v}_2 \implies \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 4 \times \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{cases} -v_1 + 2v_2 = 0 \\ v_1 - 2v_2 = 0 \end{cases} \implies -v_1 + 2v_2 = 0 \text{ for } v_1 = 2 \implies v_2 = 1$$

$$\text{Second eigen-pair is } (\lambda_2 = 4,\ \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix})$$

# Example: Eigen Decomposition of a Matrix – iii

It is

$$\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

In a general notation it is

$$\mathbf{AV} = \mathbf{V}\boldsymbol{\Lambda}$$

where $\mathbf{V}$ is the matrix of eigenvectors and $\boldsymbol{\Lambda}$ is the diagonal matrix of eigenvalues.

$\mathbf{T}$ being invertible, we can obtain

$$\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

This is the process of diagonalization, where we have transformed the matrix $\mathbf{A}$ into a diagonal matrix $\boldsymbol{\Lambda}$ using the similarity transformation defined by $\mathbf{T}$.

$$\mathbf{T}^{-1}\mathbf{AT} = \boldsymbol{\Lambda}$$

# Example: Eigen Decomposition of a Matrix – iv

**Interpretation:**

It express that the action of $\mathbf{A}$ on the eigenvectors $\mathbf{v}_i$

$$\mathbf{A}\mathbf{v}_i = \lambda_i \mathbf{v}_i$$

is simply to scale them by their corresponding eigenvalues $\lambda_i$. In other words, each eigenvector $\mathbf{v}_i$ is an invariant direction under the transformation defined by $A$, and the eigenvalue $\lambda_i$ indicates how much $A$ stretches or compresses along that direction.

The matrix equation can be written as:

$$A \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}.$$

# Distinct eigenvalues

**Fact:**

If $A$ has distinct eigenvalues, then $A$ is diagonalizable.

- **Distinct eigenvalues** means $\lambda_i \neq \lambda_j$ for $i \neq j$.
- The converse is **not** true: a matrix can have repeated eigenvalues and still be diagonalizable.

# Diagonalization

Given an $n \times n$ square matrix $\mathbf{A}$ with $n$ distinct eigenvalues.

$$\mathbf{A}\,\mathbf{T} \;=\; \mathbf{T}\,\mathbf{\Lambda}$$

If $\mathbf{T}$ is full rank (i.e., invertible), then $A$ is diagonalizable, We can obtain:

$$\mathbf{T}^{-1}\,\mathbf{A}\mathbf{T} \;=\; \mathbf{\Lambda} \;=\; \mathrm{diad}\left(\lambda_1, \lambda_2, \ldots, \lambda_n\right)$$

Hence the name **Diagonalization**.

# Not all matrices are diagonalizable

**Example:**

Given the matrix $$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

- characteristic polynomial: $P(\lambda) = \lambda^2 \quad \Rightarrow \quad \lambda = 0$ is the only eigenvalue
- eigenvectors satisfy $A\mathbf{v} = 0\mathbf{v} = \mathbf{0}$, i.e.,

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- so all eigenvectors must satisfy $v_2 = 0$, hence

$$\mathbf{v} = \begin{bmatrix} v_1 \\ 0 \end{bmatrix}, \qquad v_1 \neq 0$$

- thus, $\mathbf{A}$ has only one linearly independent eigenvector
- therefore, $\mathbf{A}$ is **not diagonalizable**
- if $\mathbf{A}$ is not diagonalizable, it is sometimes called **defective**

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# Diagonalization of Symmetric Real Matrices

Let $\mathbf{A}_{n\times n}$ be a real symmetric matrix, i.e., $\mathbf{A}^T = \mathbf{A}$. Then $\mathbf{A}$ admits an orthogonal diagonalization:

$$\mathbf{A} = \mathbf{V}\,\mathbf{\Lambda}\,\mathbf{V}^T,$$

where $\mathbf{T}^T = \mathbf{T}^{-1}$ and

$$\mathbf{T}^T\mathbf{T} = \mathbf{T}\mathbf{T}^T = \mathbf{I}.$$

Here, $\mathbf{T}$ contains the orthonormal eigenvectors of $\mathbf{A}$, and $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues.

# Similarity Transformations

- **Similarity:** If $Q$ is **square** and **nonsingular**, then

$$B = Q^{-1}AQ.$$

  In this case, $A$ and $B$ are said to be **similar** (denoted $A \sim B$).

- **Spectrum Inclusion:** If $Q$ has **full column rank** (but is not necessarily square), then every eigenvalue of $B$ is an eigenvalue of $A$:

$$\lambda(B) \subseteq \lambda(A).$$

- **Spectral Equality:** If $Q$ is **invertible** (square and nonsingular), then similar matrices share exactly the same eigenvalues:

$$\lambda(A) = \lambda(B).$$

**Takeaway:**

The matrix $Q$ acts as a **change of basis**, translating the representation of $B$ into the coordinate system of $A$ while preserving its spectral structure.

# Matrix Functions

# Matrix Functions

For a diagonalizable matrix $A = T\Lambda T^{-1}$, we can extend a scalar function $f$ (such as polynomials, exponentials, or roots) to the matrix case.

**If** $f$ is defined on the spectrum[1]of $A$, we define the **matrix function** as:

$$f(A) = T\, f(\Lambda)\, T^{-1}$$

where $f(\Lambda)$ is the diagonal matrix obtained by applying $f$ to each eigenvalue:

$$f(\Lambda) = \mathrm{diag}\left(f(\lambda_1), f(\lambda_2), \ldots, f(\lambda_n)\right)$$

[1]The *spectrum* of a matrix $A$, denoted $\sigma(A)$, is the set of all eigenvalues of $A$, i.e.,

$$\sigma(A) = \{\lambda \in \mathbb{C} : \det(A - \lambda I) = 0\}.$$

# Matrix Powers – i

**Why diagonalize?** Computing powers $A^n$ directly requires $n$ matrix multiplications. But if $A$ is diagonalizable, we can write

$$A = T\Lambda T^{-1},$$

and therefore

$$A^n = (T\Lambda T^{-1})^n = T\,\Lambda^n\,T^{-1}.$$

Since $\Lambda$ is diagonal, its powers are trivial:

$$\Lambda^n = \mathrm{diag}(\lambda_1^n, \lambda_2^n, \ldots, \lambda_n^n),$$

so the computation becomes extremely inexpensive.

# Matrix Powers – ii

**Examples:**

$$A^{1/2} = T\,\Lambda^{1/2}\,T^{-1}, \qquad e^{A} = T\,e^{\Lambda}\,T^{-1},$$

with

$$e^{\Lambda} = \mathrm{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n}).$$

**Why this matters (especially in EDA):**

Diagonalization converts difficult matrix computations into simple scalar operations on eigenvalues. This enables fast evaluation of $A^n$, $A^{1/2}$, and $e^{A}$, which arise in simulation algorithms, stability analysis, and time-domain state-transition solutions.

# Matrix Exponential – i

**Scalar exponential (review):**

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

**Matrix exponential:** For any square matrix $A$, the exponential is defined by the same power series:

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}.$$

**Key idea:** The series converges for every matrix $A$, and $e^A$ behaves like a generalization of the scalar exponential to matrices.

**1. Power–series definition:**

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}.$$

# Matrix Exponential – ii

**2. Diagonalization:** If $A$ is diagonalizable,

$$A = T\Lambda T^{-1},$$

where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

**3. Put the series into the diagonal form:**

$$A^n = (T\Lambda T^{-1})^n = T\,\Lambda^n\,T^{-1}.$$

**4. Substitute into the series:**

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!} = \sum_{n=0}^{\infty} \frac{T\Lambda^n T^{-1}}{n!} = T\left(\sum_{n=0}^{\infty} \frac{\Lambda^n}{n!}\right) T^{-1}.$$

**5. But $\Lambda$ is diagonal, so**

$$e^{\Lambda} = \mathrm{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n}),$$

and therefore

$$\boxed{e^A = Te^{\Lambda}T^{-1}}.$$

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